Cambridge (CIE) A Level Computer Science

Your notes

Protocols

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- Protocol Examples

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Need for protocols

What is a protocol?

- A protocol is a set of rules that define how computers **communicate over a network**
- They ensure that data is **sent**, **received**, and **understood** correctly by all devices, regardless of their manufacturer or operating system
- Without protocols:
  - Devices would **not know how** to structure or interpret data
  - Data could be **lost**, **corrupted**, or **misrouted**
  - Communication between **different hardware and software** systems would break down
- **Protocols are essential** to guarantee:
  - Reliable data transfer
  - Consistent formatting and addressing
  - Error checking and correction
  - Security during transmission
- **Examples** of protocols include:
  - TCP, IP, HTTP, FTP, SMTP

Protocol stacks

- The protocol stack is a way of **organising network protocols into layers**, where each layer is responsible for a specific function
  - Each layer **handles its own task independently**
  - Layers **communicate with the layers directly above and below**
  - The **stacked model** makes the network easier to design, update, and troubleshoot
- Makes communication **modular** and **efficient**
- Allows different technologies to **work together** using standardised rules
- An example of a protocol stack is **TCP/IP**

TCP/IP

What is TCP/IP?

- TCP/IP, or the Transmission Control Protocol/Internet Protocol, is a **suite of communication protocols** used to interconnect network devices on the internet

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- This model splits the various protocols into four layers:
  - **Application**
  - **Transport**
  - **Internet**
  - **Link**

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Application layer

- This is the layer where the **communication process begins**
- The application layer **interacts directly with software applications**, such as web browsers and email clients
- The application layer **prepares data for transmission** over the network by converting it into a format that can be sent and received over the network (known as encapsulation)

Transport layer

- The transport layer **receives data from the application layer**
- The transport layer is **responsible for end-to-end communication** between the source and destination
- The transport layer **breaks the data it receives down into smaller units** called packets
- Each packet is **assigned a port number** (so the data can be reassembled in the correct order at the destination)
- Each packet is also **labelled with a header** containing information (e.g. the packet number)

Internet layer

- The internet layer **receives packets from the transport layer**
- It **adds a header to each packet**, including the sender's **IP address** and the receiver's IP address
- The internet layer is responsible for **routing each packet across the network** using the IP addresses in the headers

Link layer

- Also known as the **network interface layer**
- The link layer **receives packets from the Internet layer** and prepares them for transmission over the physical network
- The link layer **translates the digital packets** into an electrical, optical, or wireless signal that can be sent over the network
- Once the signal reaches the receiving end, the network layer **translates it back** into digital packets

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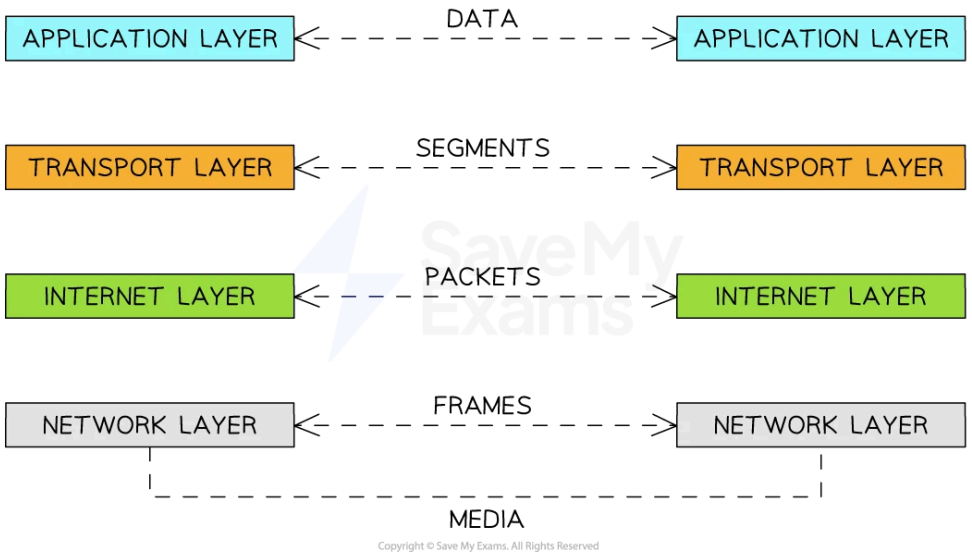
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TRANSPORT LAYER
SEGMENTS
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INTERNET LAYER
PACKETS
INTERNET LAYER
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Protocol Examples

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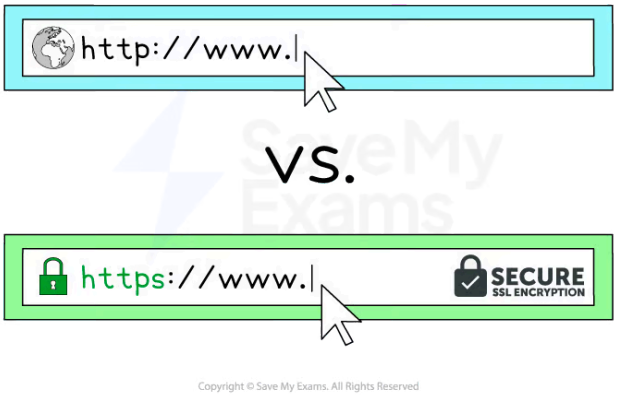
Standard protocols

- A protocol is a **set of rules** that govern **communication** on a network
- There are protocols for different purposes, such as:
  - **HTTP & HTTPS**
  - **FTP**
  - **POP, IMAP & SMTP**
  - **BitTorrent**

HTTP & HTTPS

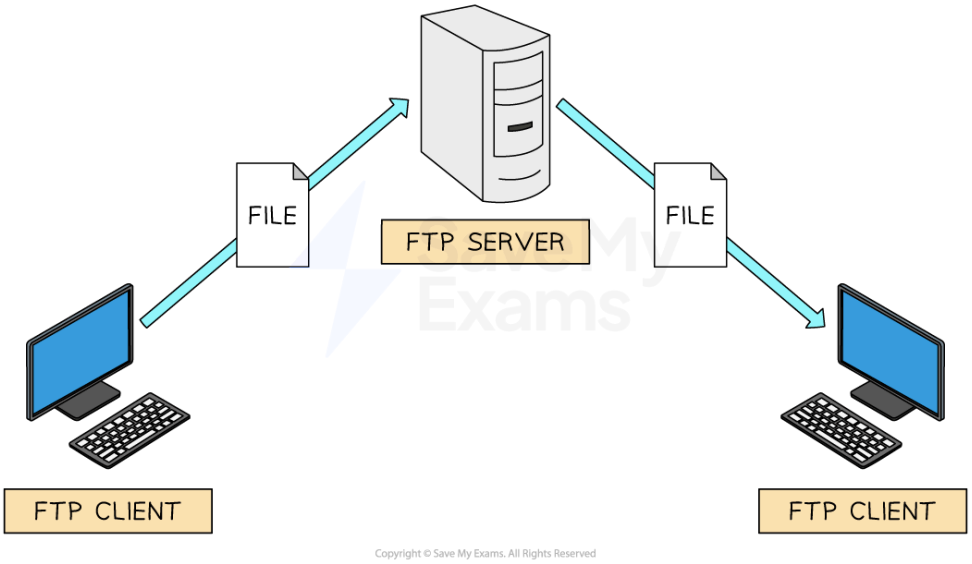
- Hypertext Transfer Protocol (**HTTP**) allows communication between clients and servers for **website viewing**
- HTTP allows clients to **receive data** from the sever (fetching a webpage) and **send data** to the server (submitting a form, uploading a file)
- **HTTPS** works in the same way as HTTP but with an **added layer of security**
- All data sent and received using HTTPS is **encrypted**
- HTTPS is used to **protect sensitive information** such as passwords, financial information and personal data



FTP

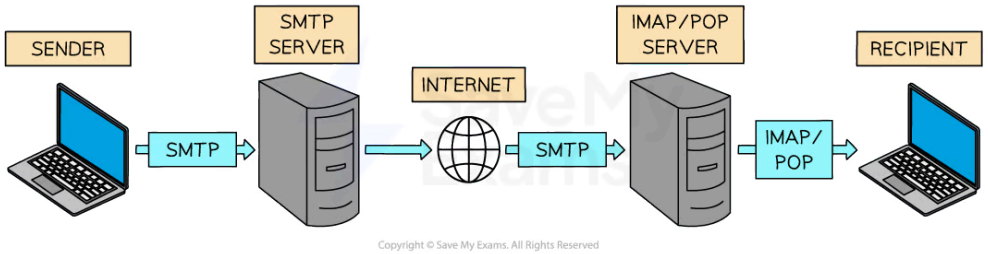
- File Transfer Protocol (**FTP**) allows **sending and receiving files between computers**
- Uploading and downloading files **to/from a web server** is often completed using FTP
- FTP offers **greater efficiency** and support for **bulk transfers and large files** such as resuming interrupted transfers

- **FTP clients** are software applications that use the FTP protocol to make the process easier for users



POP, IMAP & SMTP

- A family of protocols that handle **sending and receiving of email** across the internet (WAN)



SMTP

- Simple Mail Transfer Protocol (**SMTP**) is a protocol that allows **communication** between an **email sender** and the **email server**, and between different email servers using the internet
- In the diagram above, **SMTP** is used to transfer the senders email to their email providers server and SMTP is used to transfer the email to the recipients email server

POP

- Post Office Protocol (**POP**) is a protocol for **downloading emails to a device** from an email server
- Once the email has been retrieved **it is removed from the server**

IMAP

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- Internet Message Access Protocol (**IMAP**) is a protocol for **downloading emails to a device** from an email server
- Once the email has been retrieved, **a copy is retained on the mail server**

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Advantages & disadvantages of POP/IMAP

	Advantages	Disadvantages
POP	▪ Frees up storage space on email servers ▪ Faster on slow connections	▪ Only access emails from the device they're downloaded to ▪ Emails deleted on the server once downloaded
IMAP	▪ View and manage emails from any device with internet access ▪ Changes made on one device are synchronised on all connected devices	▪ Server storage space can limit amount of retained emails ▪ Requires internet access to view emails

BitTorrent

- BitTorrent is a **peer-to-peer (P2P) file-sharing protocol** used to **distribute large amounts of data** efficiently across the internet
- Instead of downloading a file from a single central server, BitTorrent allows users to **download pieces of the file from multiple users (peers)** who already have parts of it
- This makes file sharing:
  - **Faster**, as downloads come from many sources at once
  - **More efficient**, as it reduces the load on any single server
- BitTorrent is often used for:
  - Sharing **large files**, such as software, videos, or games
  - **Decentralised distribution**, which avoids the need for central hosting
- While BitTorrent is a legal technology, it is sometimes used for sharing **copyrighted content illegally**, so ethical and legal use is important